CERFACS

Quantum Hackathon:

Fluid and Spatial problems

Meteo and Climate Quantum Simulation with QUOBLY

[October 26, 2026]



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Silicon Spin Qubits

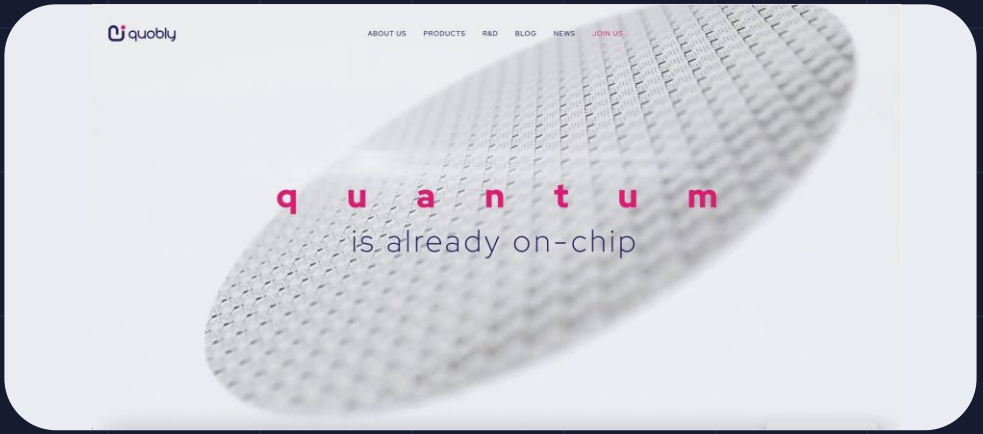
SpinPulse

Karla Baumann

Fluid Dynamics



Quantum Solutions



quobly

Silicon-based

Quantum Computers

built for existing infrastructures

Discover Alloy Series



Quobly in 2026



€10M

Revenue
First commercial contract through PROQCIMA



65
Patents families

120+
People

x3
By mid 2028

54%
PhDs

18
Nationalities

5 Locations

Grenoble FR
(HQ) Paris FR
Sherbrooke CA
Singapore
Taiwan

20+
partnerships



Exclusive partnership to develop our proprietary QSOI® qubit technology and produce million-qubit chips

June 2026: Quobly secures 115M€

Reservoir Computing

Input $\mathbf{x}(t)$

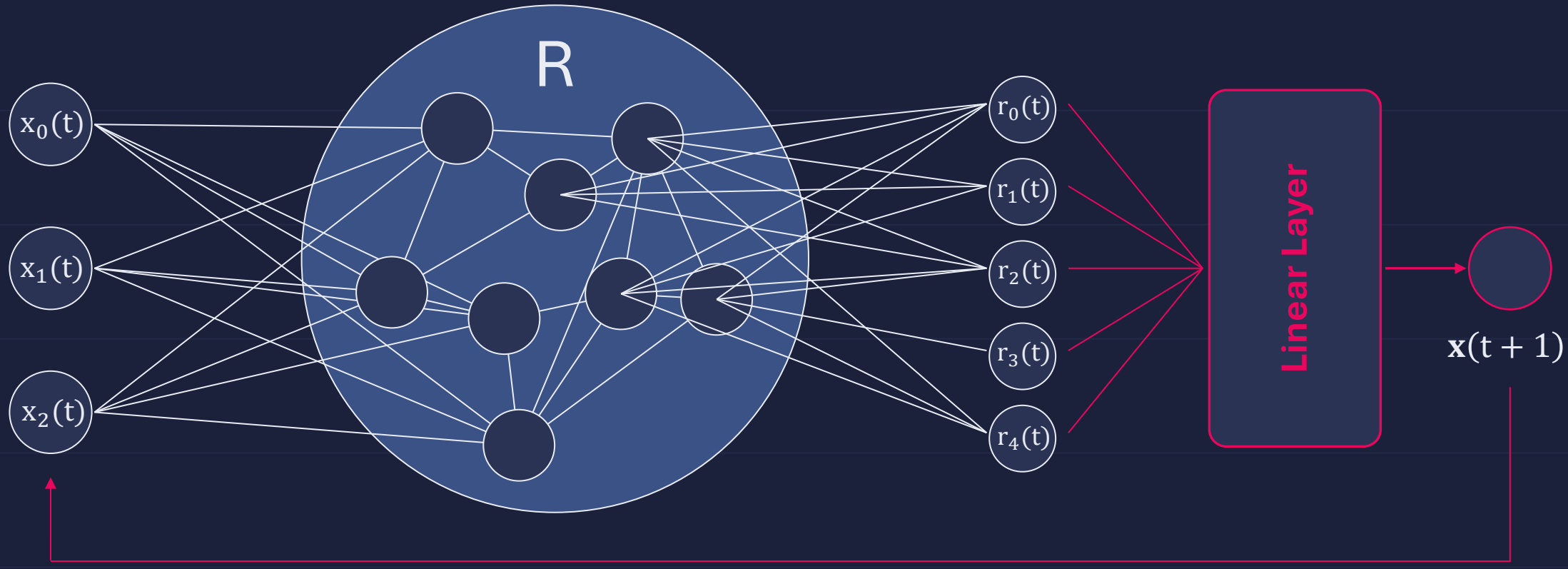
Reservoir

Randomly initialized and never trained

Measurement

Linear Layer

The only trained part, always fully classical



Why QRC now

Reservoir computing

- Cheap to train: random fixed reservoir, linear readout only
- Classification, anomaly detection, but built for time-series forecasting
- No backpropagation through time

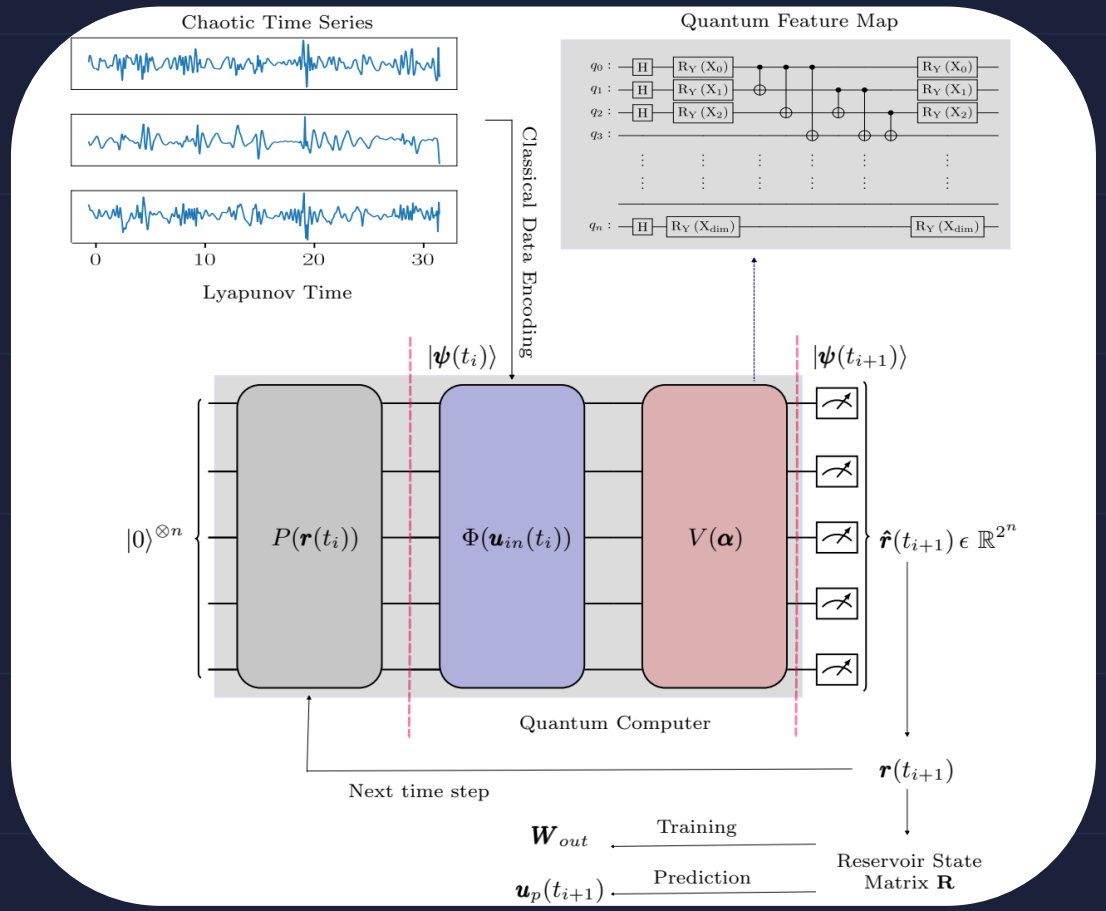
Quantum RC

- Dimension scales as 2^n; 10 qubits ≈ 1024 features
- Entanglement gives cross-terms between inputs for free
- Circuit stays untrained: no gradients on hardware, no barren plateaus

QRC now

- A real NISQ application: no optimization loop, and the right noise even helps (dissipation supplies the fading memory)
- No minimum qubit count to be useful
- Already run on real hardware

Quantum Reservoir Computing (QRC) on Quantum Circuits



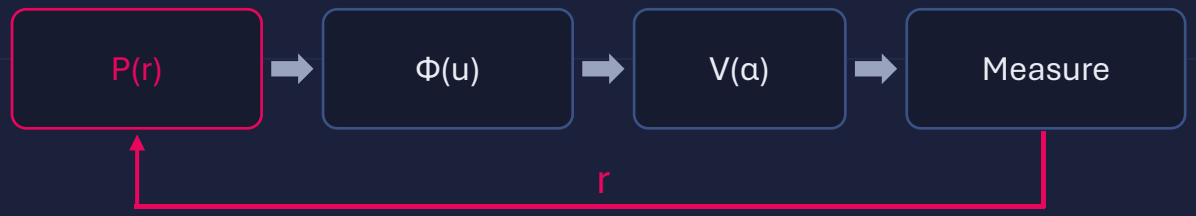
QRC with recurrence

1. *Encode the previous reservoir state into the circuit*
2. *Encode the data — the next datapoint during training, the last prediction during forecasting*
3. *Apply a fixed random circuit evolution*
4. *Measure the qubits to read the reservoir state*
5. *Train a ridge regression on the measured states*

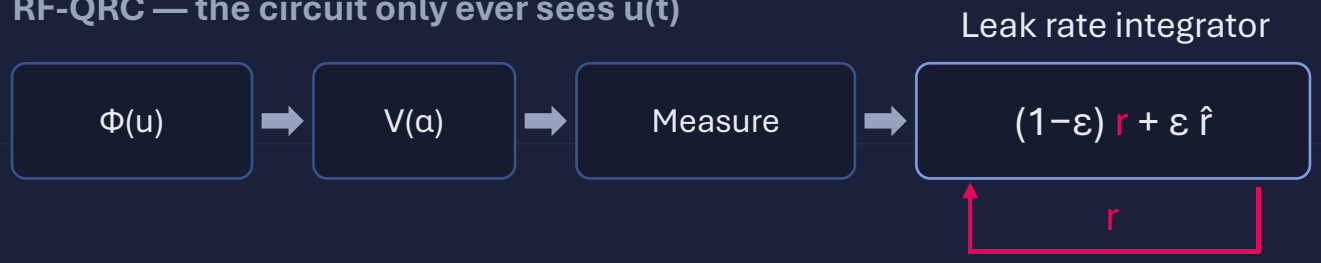
"Prediction of chaotic dynamics and extreme events: A recurrence-free quantum reservoir computing approach" by Osama Ahmed, Felix Tennie, Luca Magri, submitted 6 May 2024, arXiv:2405.03390

Recurrence-free QRC (RF-QRC)

QRC — with recurrence



RF-QRC — the circuit only ever sees u(t)



RF-QRC advantages

- *No reservoir state fed back into the circuit, each step encodes only the current input*
- *The feedback moves outside, into a classical leaky integrator whose leak rate ε sets how long the past survives*
- *Sampling noise has less impact, no longer propagates from step to step*
- *Every circuit evaluation is independent, so training runs in parallel— only the closed-loop forecast need to be sequential*

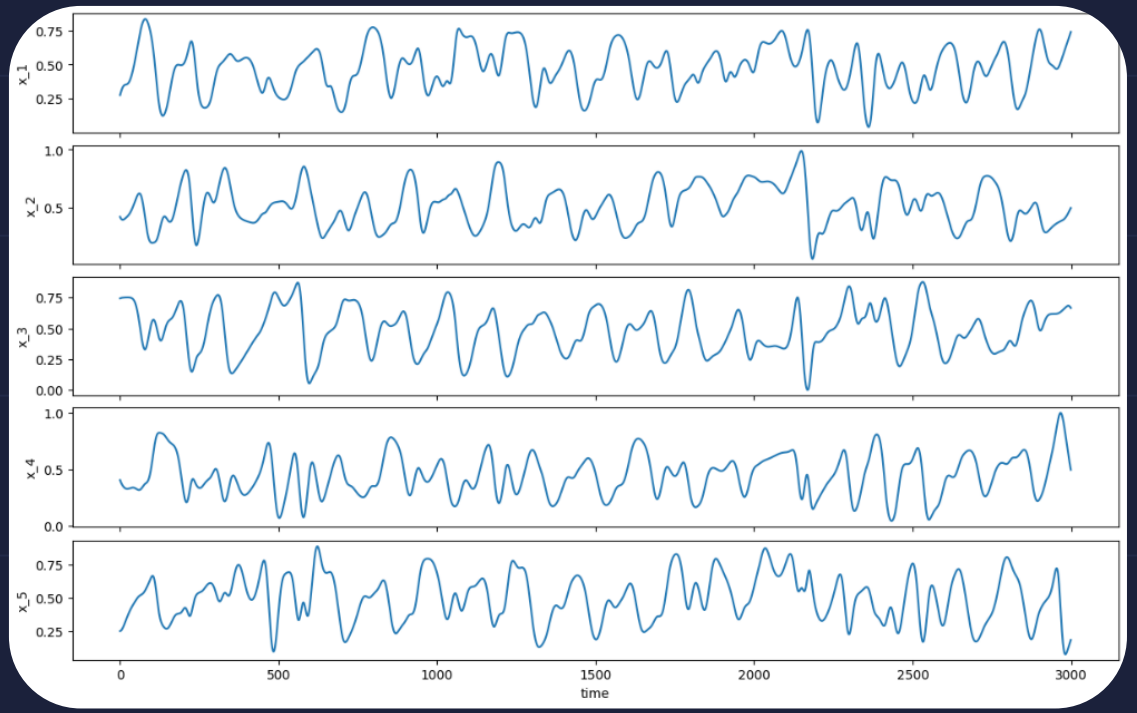
Close to a "Quantum Extreme Learning Machine", but with a classical memory.

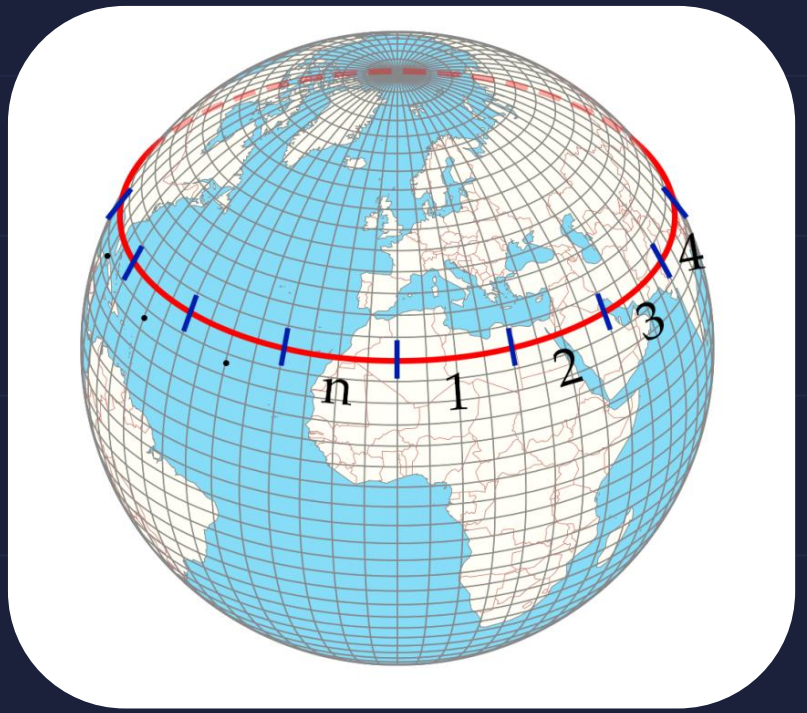
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Lorenz 96

$$\frac{dX_k}{dt} = X_{k-1}(X_{k+1} - X_{k-2}) - X_k + F$$

A standard chaotic benchmark: n sites on a latitude circle, dimension set by n, each carrying an atmospheric quantity propagating westward.





Multiscale Lorenz-96

"Data-driven prediction of a multi-scale Lorenz 96 chaotic system using a hierarchy of deep learning methods: Reservoir computing, ANN, and RNN-LSTM" by Ashesh Chattopadhyay, Pedram Hassanzadeh, Krishna V. Palem, Devika Subramanian, submitted 20 June 2019, arXiv:1906.08829

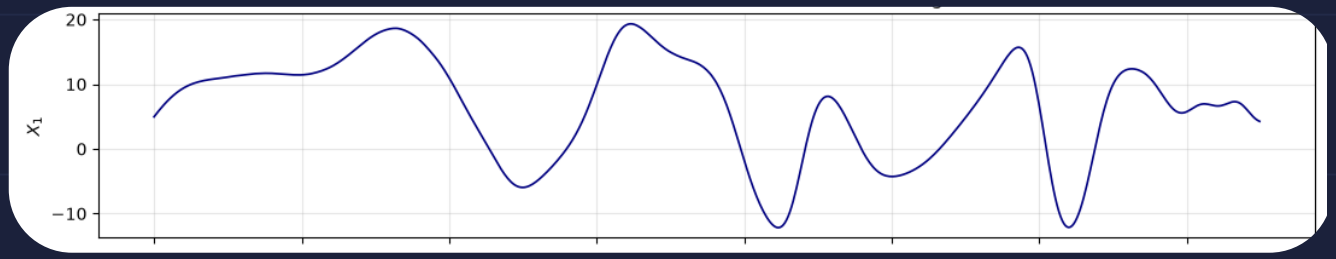
$$\frac{dX_k}{dt} = X_{k-1}(X_{k+1} - X_{k-2}) - X_k + F - \frac{hc}{b}\sum_j Y_{j,k}$$

$$\frac{dY_{j,k}}{dt} = -cb\, Y_{j+1,k}(Y_{j+2,k} - Y_{j-1,k}) - c\, Y_{j,k} + \frac{hc}{b} X_k - \frac{he}{d}\sum_i Z_{i,j,k}$$

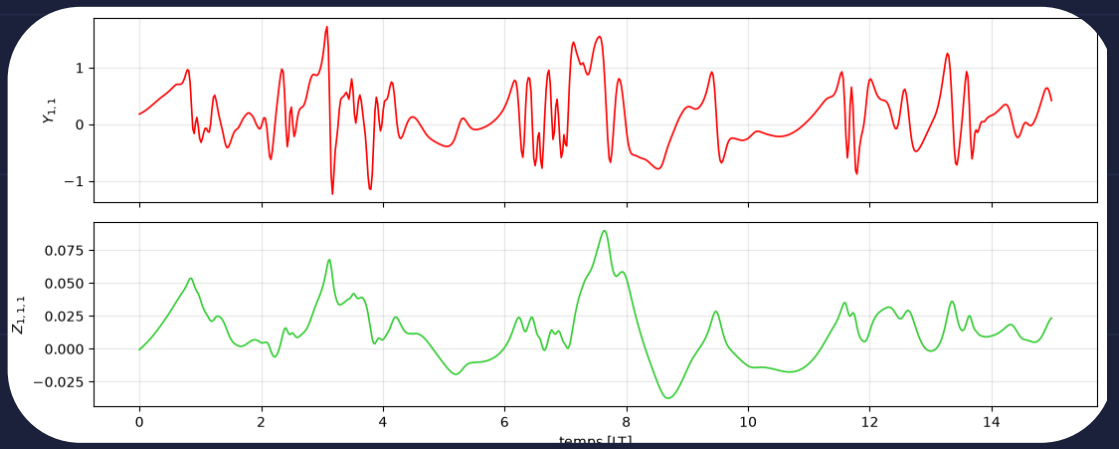
$$\frac{dZ_{i,j,k}}{dt} = ed\, Z_{i-1,j,k}(Z_{i+1,j,k} - Z_{i-2,j,k}) - g\, e\, Z_{i,j,k} + \frac{he}{d} Y_{j,k}$$

Only X is observed; the fast scales Y and Z act as unknown influences, and that is the missing physics the reservoir has to learn.

X_i : the data we want to predict



Y_i, Z_i : Unmeasured high frequency influences



- Quantum-based simulation · CFD and PDEs

Where each method enters the weather forecast

quobly

01
Data assimilation
Observations → best estimate of today's state

02
Forecast model
Physics plus parameterized small scales

03
Ensemble
~50 perturbed runs → forecast uncertainty

04
Products
Probabilities, warnings, user forecasts

QUANTUM RESERVOIR COMPUTING · NISQ

Learns to step the forecast forward

How: Trained on past model or reanalysis trajectories, the reservoir predicts the next state, $X(t) \rightarrow X(t + \Delta t)$. Each prediction feeds the next step.

Weather analogue: data-driven emulators such as ML weather models, or a learned parameterization of unresolved scales.

KOLMOGORIAN ALGORITHM · FTQC

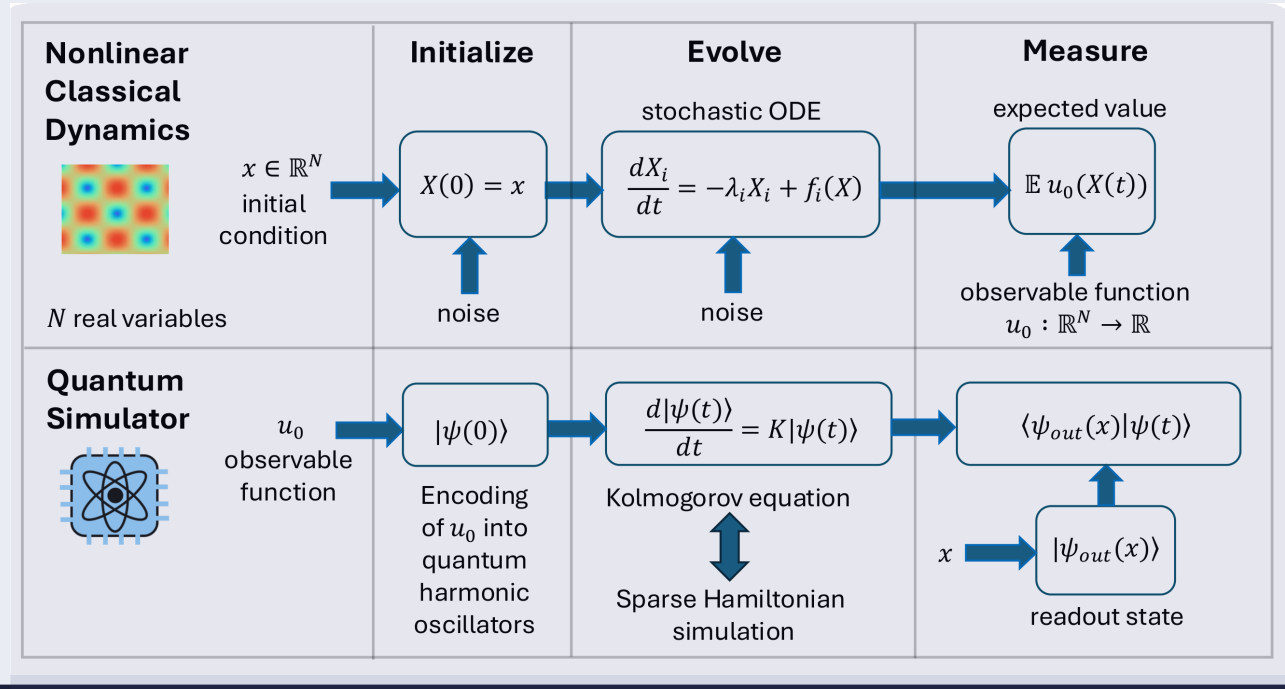
Computes ensemble means directly

How: Stochastic parameterization turns the forecast model into a stochastic differential equation; the algorithm returns the ensemble mean $E[X(t)]$ of a chosen quantity without running members.

Weather analogue: the expected value of one variable at one location in a stochastic-physics ensemble.

Non-linear dynamics, turbulence, and Navier–Stokes

> « *Here we show that* ***strongly nonlinear (R≫1)*** *classical dynamics with a quadratic nonlinearity can be* ***efficiently simulated on a quantum computer*** *provided that the* ***simulated system is both dissipative and noisy***. »



Efficient quantum simulation of strongly nonlinear stochastic dynamics

- New Kolmogorov-equation embedding and regularization framework
- Poly(log N), polynomial-in-t quantum algorithm for observables
- Applicable to turbulence models (2D Navier–Stokes vortex flow)
- Strong evidence for a quantum advantage / BQP-completeness

[Bravyi et al., 2025] arXiv:2507.06198

Two quantum routes to a weather forecast

	Quantum reservoir computing NISQ · runs on today's hardware	**Kolmogorian algorithm** FTQC · needs fault-tolerant hardware
Role in forecasting	Forecast model: emulator or learned parameterization	Ensemble: forecast statistics without sampling
Output	Point forecast of the state	Ensemble means and low-order moments
Hardware	Today's noisy devices, no error correction	Error-corrected machines, not yet available
Quantum resource	8 qubits, fixed shallow circuits, ~10^4 shots per circuit	Block encodings, long Hamiltonian simulation, Hadamard test
Guarantee	None: heuristic machine learning	Proven error bounds; cost polylog(N), linear in time
Main limit	Shot noise; sequential steps, minutes to hours per forecast	Constant forcing excluded; cost exponential in anomaly size

Kolmogorian algorithm: Bravyi, Byrne, Zayats, Zhuk (2026), arXiv:2606.08349; Bravyi, Manson-Sawko, Zayats, Zhuk (2025), arXiv:2507.06198.

NISQ today, fault tolerance later

NISQ · AVAILABLE TODAY

Noisy qubits, no error correction

Hundreds of physical qubits; every gate adds error, so only shallow circuits run reliably.

Fits quantum reservoir computing

One fixed, shallow circuit per step; noise is tolerated as part of the random reservoir; training is classical ridge regression.

error correction

FTQC · FUTURE HARDWARE

Logical qubits, deep circuits

Error correction builds logical qubits from many physical ones, so long, deep circuits become reliable.

Needed for the Kolmogorian algorithm

Block encodings, long Hamiltonian simulation and high-precision overlap estimates need deep, coherent circuits.

Complements, not rivals: QRC can test a learned forecast model on today's hardware; the Kolmogorian algorithm targets ensemble statistics once machines are fault tolerant.

SpinPulse

*An open-source Python package from Quobly: it simulates spin qubit circuits at the **pulse level**, not the gate level.*

Gate-level simulation

- *Gates are fixed matrices*
- *One Kraus channel per gate, drawn independently*
- *The environment forgets between gates*

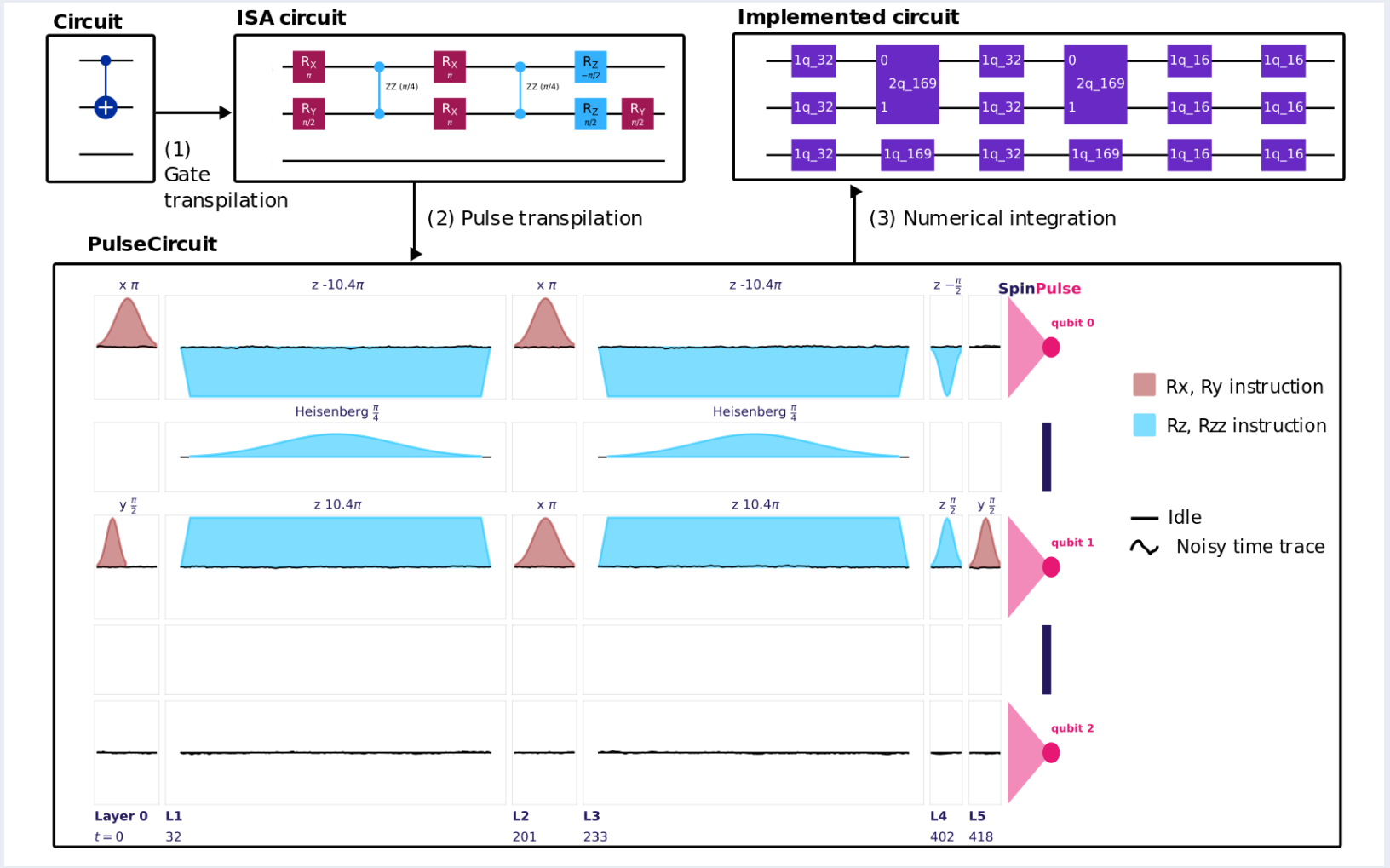
SpinPulse — pulse level simulation

- *Gates are control signals with a duration*
- *The device Hamiltonian is integrated under them*
- *One noise trace covers the whole circuit*

SpinPulse allows non-Markovian noise simulation, which is not possible with gate-level simulations.

"The SpinPulse library for transpilation and noise-accurate simulation of spin qubit quantum computers" by Benoît Vermersch, Oscar Gravier, Nathan Miscopein, Julia Guignon, Carlos Ramos Marimón, Jonathan Durandau, Matthieu Dartiailh, Tristan Meunier, Valentin Savin, submitted 15 January 2026, arXiv:2601.10435

Workflow



(1) Gate transpilation

Rewritten into the native gate set: RX, RY, RZ, RZZ.

(2) Pulse transpilation

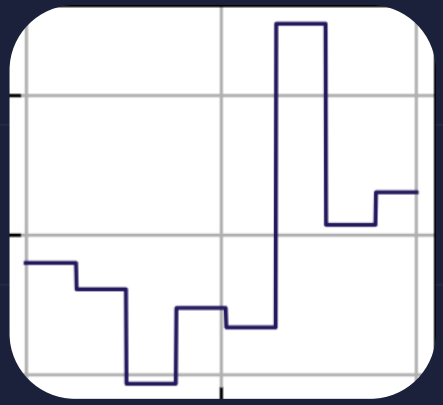
Each gate becomes a signal with a duration; idle times synchronize the qubits.

(3) Numerical integration

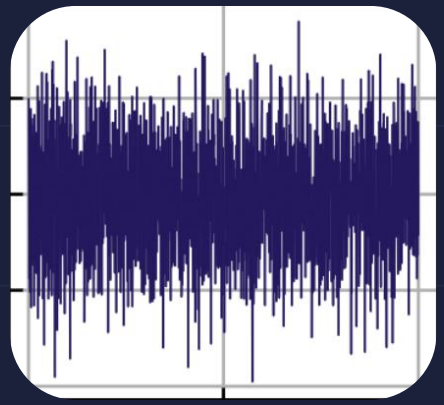
The Hamiltonian is integrated step by step, noise included.

Noise integration

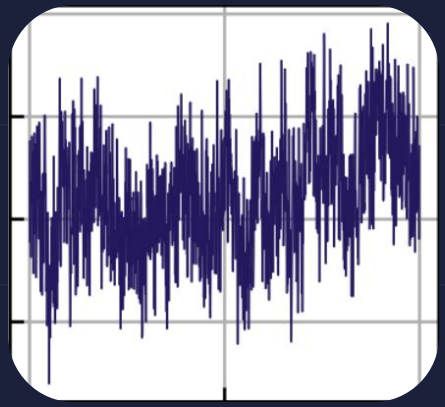
Three noise types are available: quasi-static, white and **pink**, *the 1/f noise that dominates in spin qubits.*



Quasi-static noise



White noise



Pink noise

1 — Draw a trace

A noise trace ε(t) is generated for the whole circuit, with the spectrum of the chosen type.

2 — Simulate on it

The circuit is integrated on that trace: every gate sees a consecutive slice of the same ε(t).

3 — Average

A new trace is drawn and the process repeats. Fidelities are averaged over all realizations.

Pink noise has more weight at low frequencies: it drifts over many gates, so their errors are correlated.

Alloy forge and qBraid

Run it in the cloud

- qBraid — cloud notebooks with real QPUs and simulators
- 100 h of free CPU time per month
- Alloy Forge — simulator of Quobly's Pioneer QPU
- Up to 10 qubits, free
- Based on SpinPulse code, with Quobly’s hardware specs already set



You can use Alloy Forge for free, directly on qBraid or by sending your code from your own laptop with an API key.